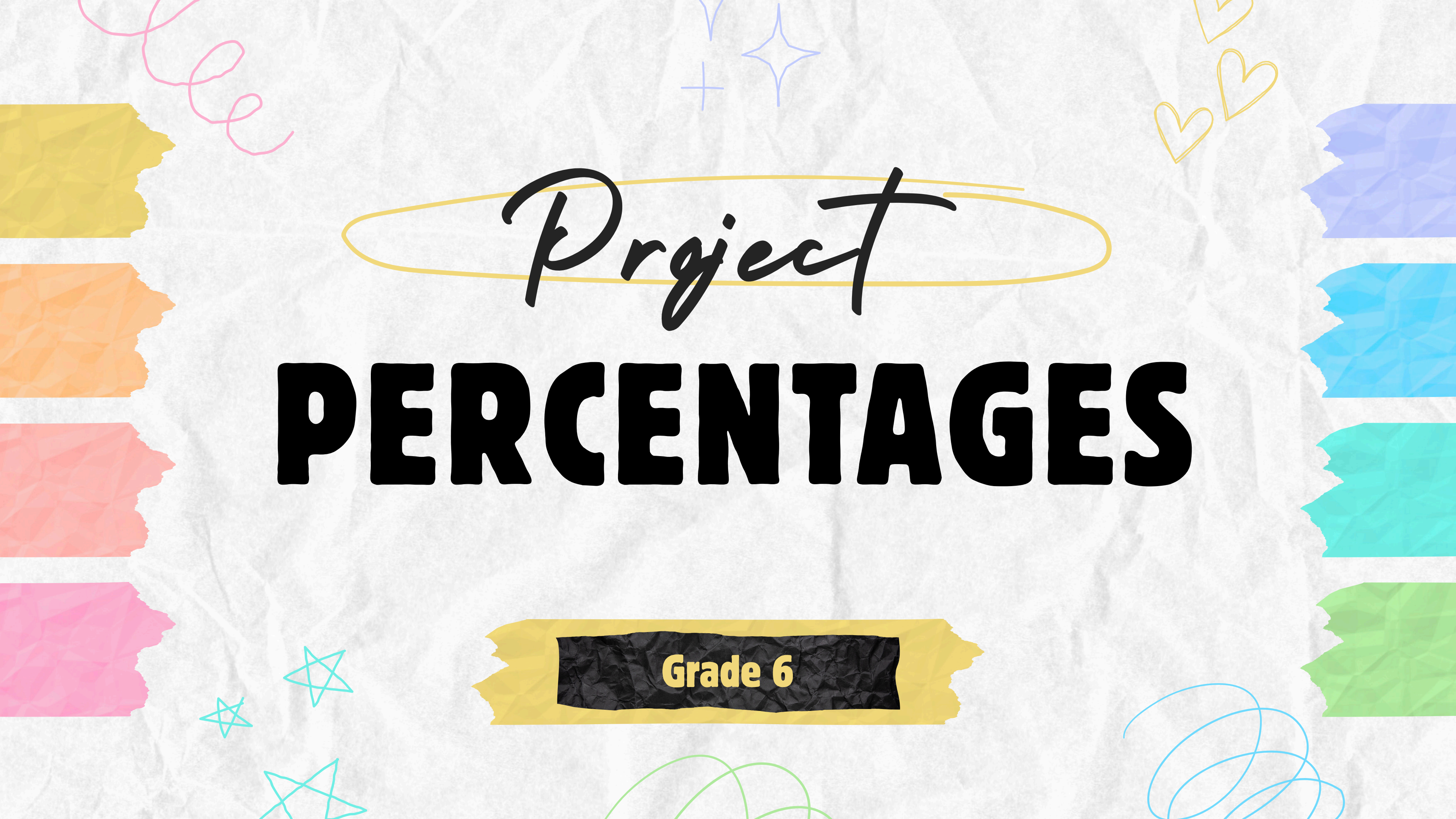




Project

PERCENTAGES

Grade 6

Lesson Outline

Understand what percentages are



Convert fractions into percentages



Express a quantity as a percentage of another (calculator)

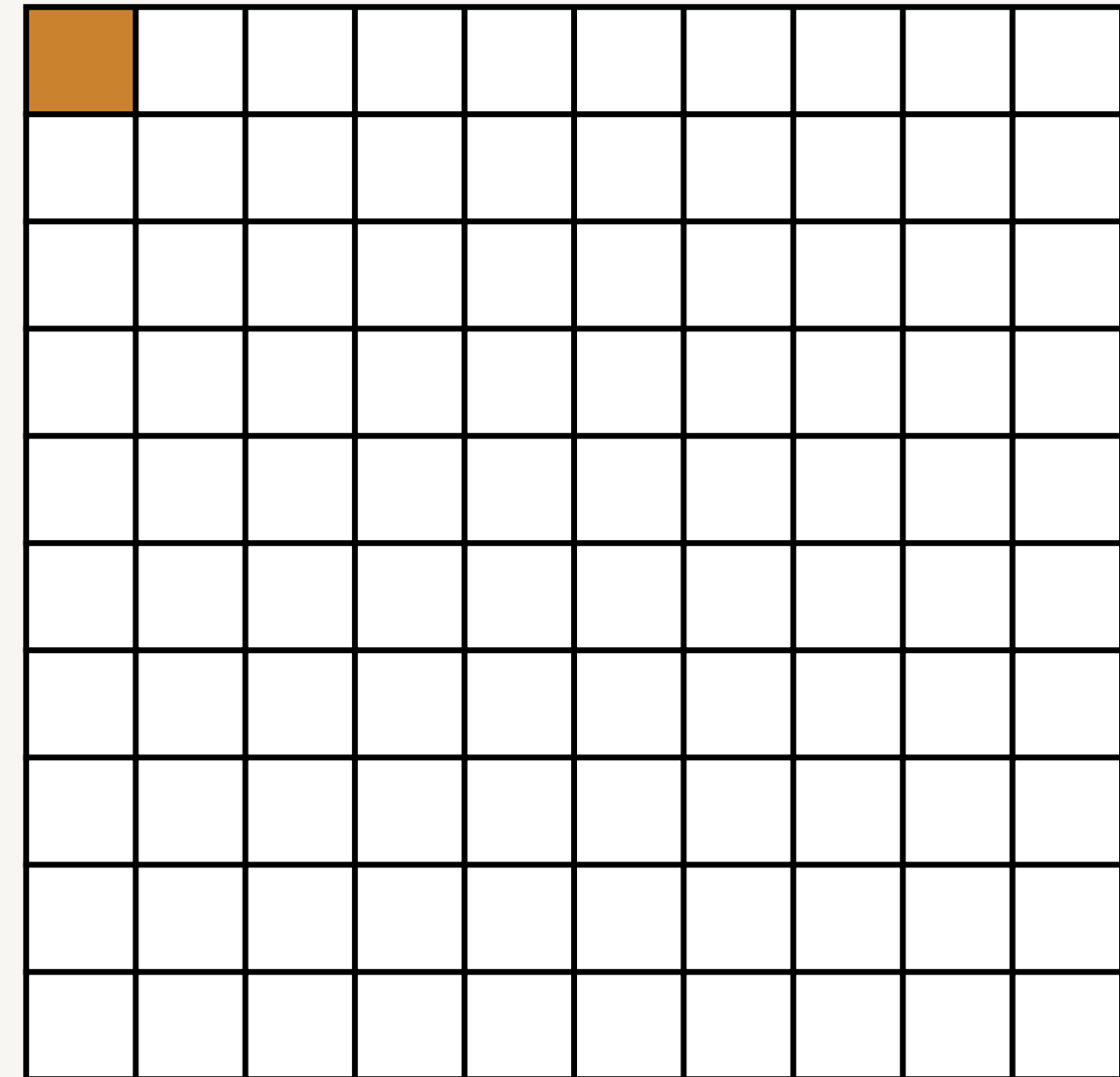


Express a quantity as a percentage of another (non - calculator)

What are Percentages?

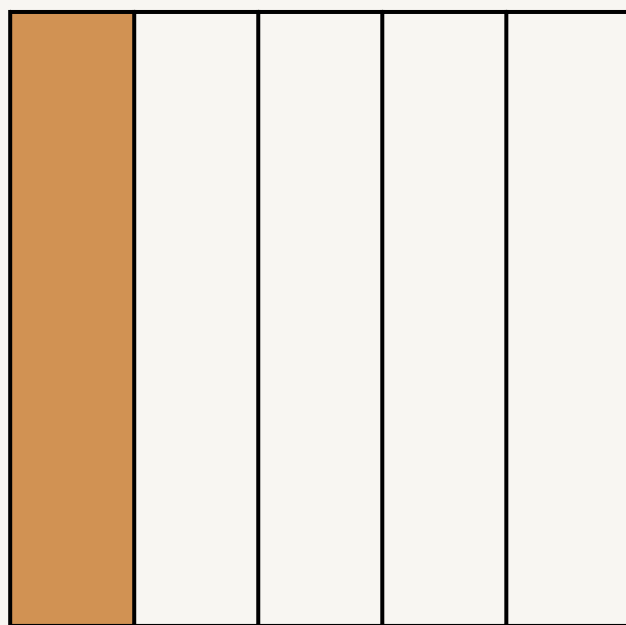
Percentages represent the number of parts per hundred.

$$\frac{1}{100} = 1\%$$

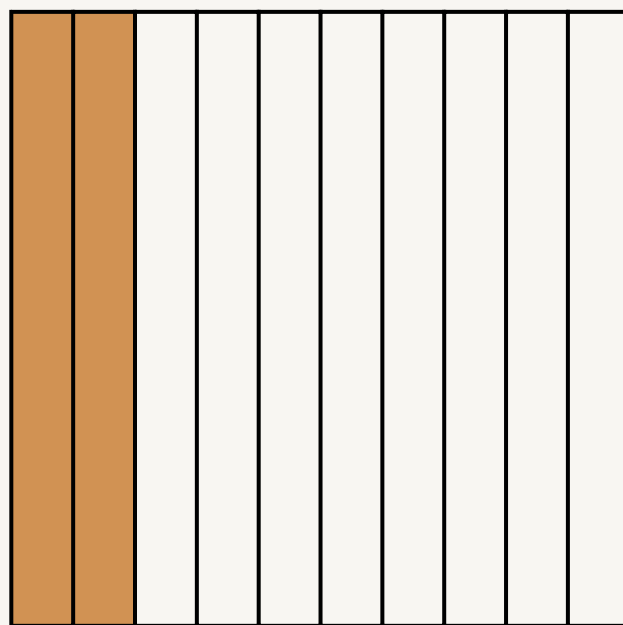


Fractions to Percentages

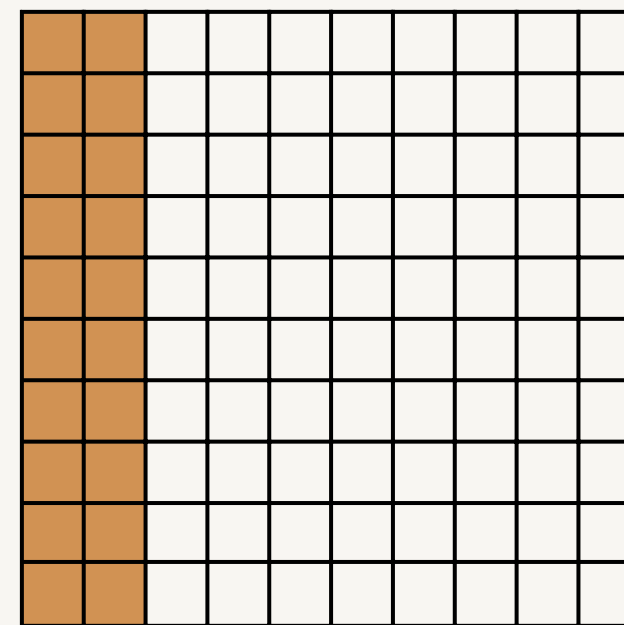
To convert fractions into percentages, you need to find an equivalent fraction where the denominator is 100, the numerator represents the percentage.



$$\frac{2}{5}$$



$$\frac{4}{10}$$



$$\frac{40}{100} = 40\%$$

Expressing one quantity as a percentage of another

**Express 16 out of 50
as a percentage**



1. Write 16 out of 50 as a fraction

$$\frac{16}{50}$$

2. Find the equivalent fraction with a denominator of 100.

$$\frac{16}{50} = \frac{32}{100}$$

3. The numerator is the percentage

$$32\%$$

Practice Questions

Express the first quantity as a percentage of the second



1. 42 out of 60

2. 33 out of 150

3. 12 minutes out of an hour

4. 600g out of 2kg

Solutions

Express the first quantity as a percentage of the second



1. 42 out of 60

$$\frac{42}{60} = \frac{7}{10} = \frac{70}{100}$$

70%

2. 33 out of 150

$$\frac{33}{150} = \frac{11}{50} = \frac{22}{100}$$

22%

3. 12 minutes out of an hour

$$\frac{12}{60} = \frac{2}{10} = \frac{20}{100}$$

20%

4. 600g out of 2kg

$$\frac{600}{2000} = \frac{60}{200} = \frac{30}{100}$$

30%

Expressing one quantity as a percentage of another

**Express 9 out of 40
as a percentage**



1. Write 9 out of 40 as a fraction

$$\frac{9}{40}$$

2. Multiply the fraction by 100

$$\frac{9}{40} \times 100$$

3. The answer is the percentage

$$\frac{9}{40} \times 100 = 22.5\%$$

Practice Questions

Express the first quantity as a percentage of the second



1. 68 out of 85

2. 29 out of 70

3. 21 seconds out of 2 minutes

4. 150 metres out 2 kilometres

Practice Questions



Express the first quantity as a percentage of the second

1. 68 out of 85

$$\frac{68}{85} \times 100 = 80\%$$

80%

2. 29 out of 70

$$\frac{29}{70} \times 100 = 41.4\%$$

41.4%

3. 21 seconds out of 2 minutes

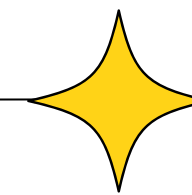
$$\frac{21}{120} \times 100 = 17.5\%$$

17.5%

4. 150 metres out 2 kilometres

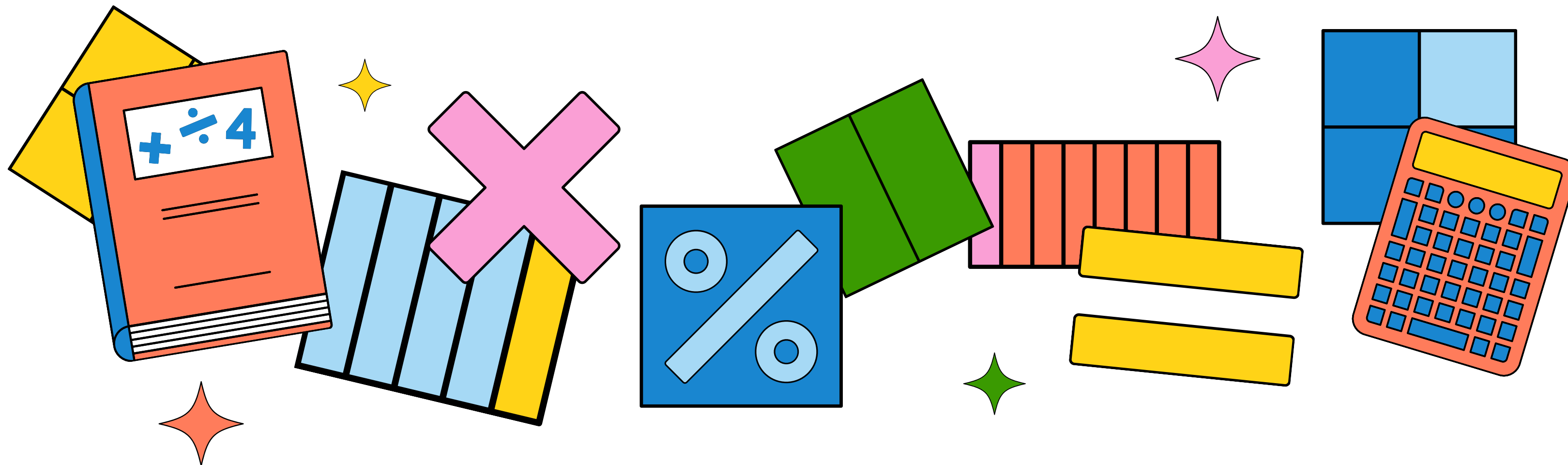
$$\frac{150}{2000} \times 100 = 7.5\%$$

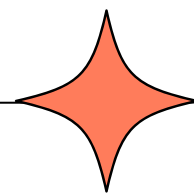
7.5%



Percentage of a Quantity

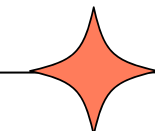
Year 6 Mathematics





50% of Year 6 students are boys. If there are 200 Year 6 students, how many are boys?

Let's find out as we go along the lesson.

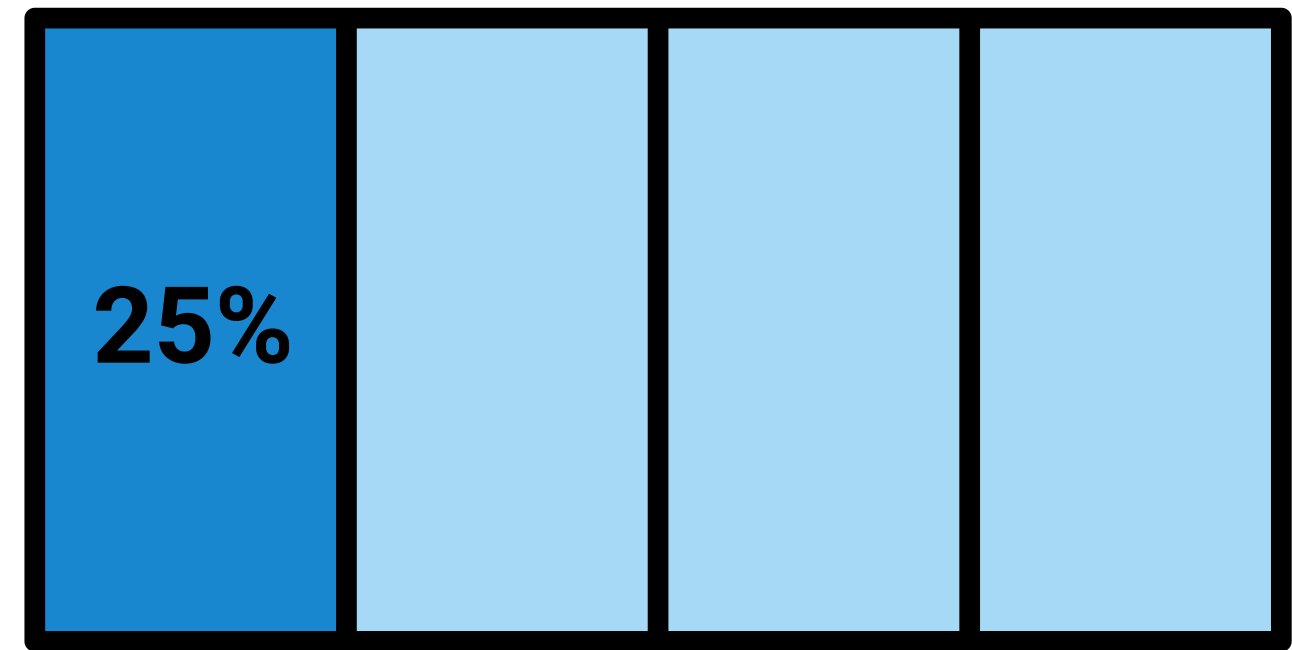


Percentage of a Quantity

▶▶ Percent means part per one hundred.

25% of 20

means getting $\frac{25}{100}$ or $\frac{1}{4}$ of 20.



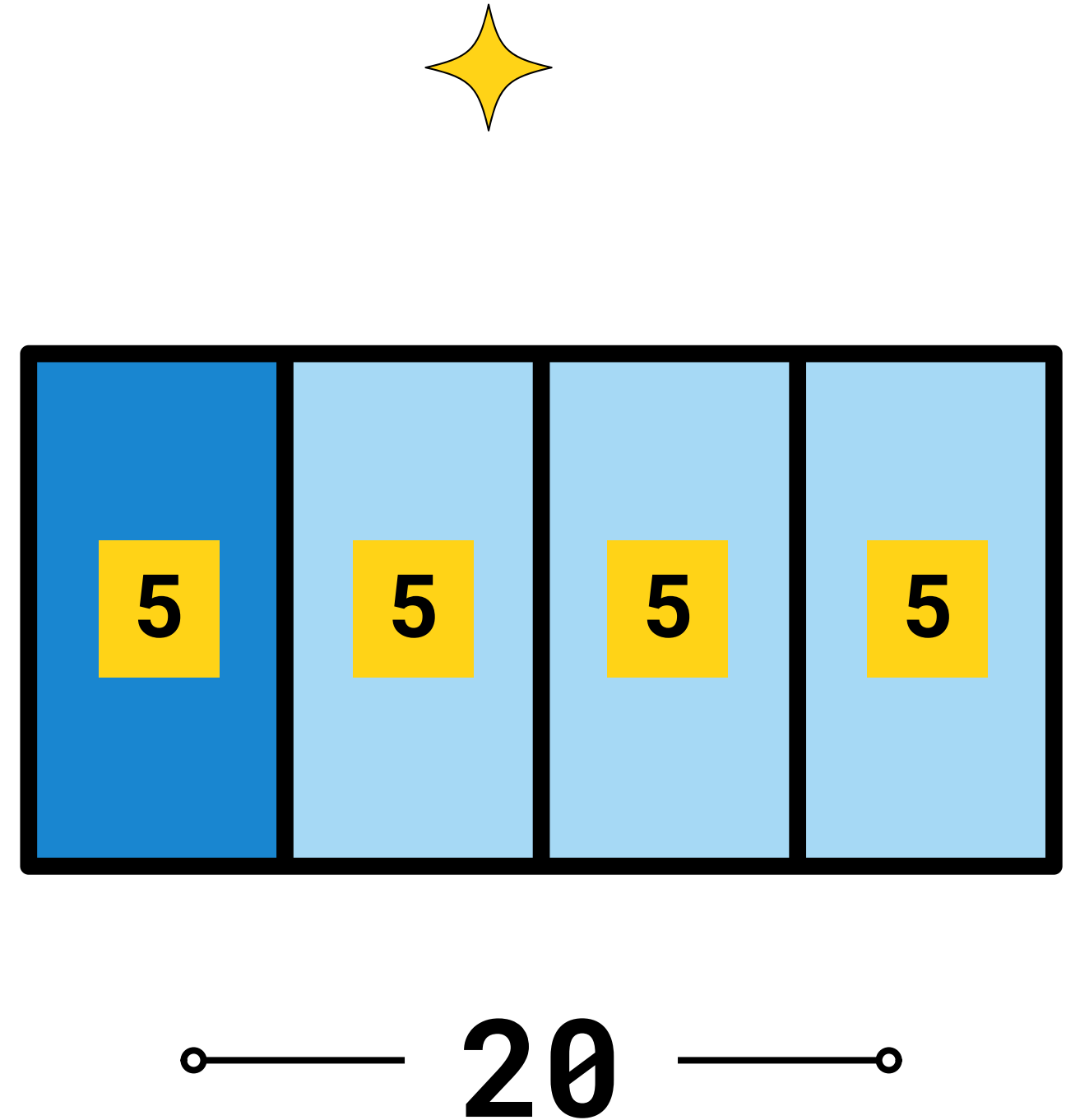
○ — **20** — ○

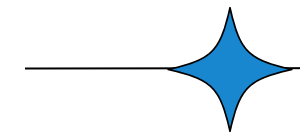
Percentage of a Quantity

▶▶ Percent means part per one hundred.

$$\frac{1}{4} \text{ of } 20 = 5$$

Breaking 20 into fourths, we'll have 5 per part.

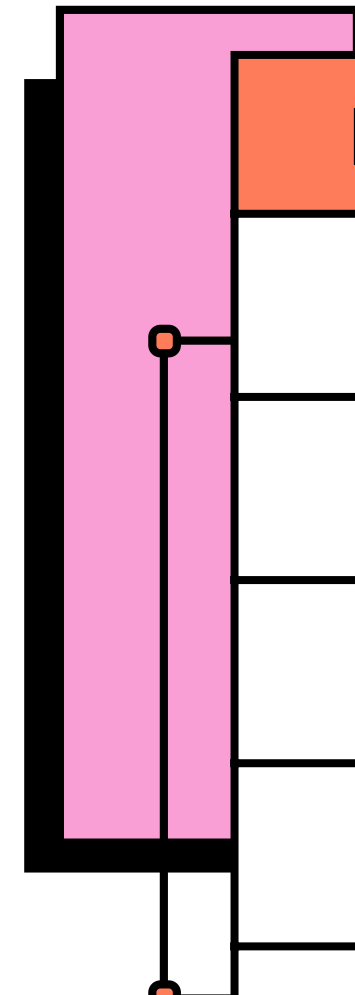




Percentage as Fractions

We can easily find the fraction equivalent of a percentage by dividing the percentage value by 100.

▶▶ **Let's look at some common equivalences.**



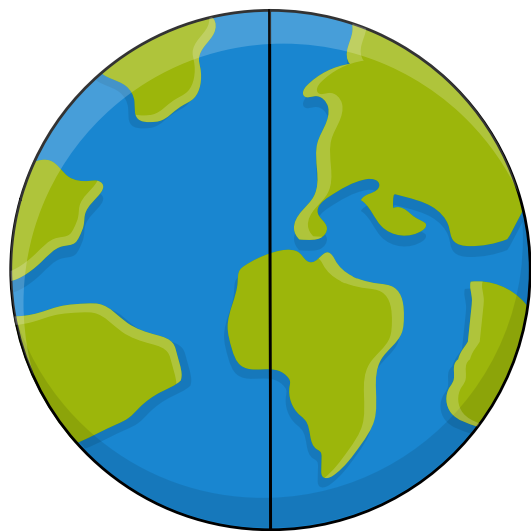
Percentage	Fraction
100%	1/1
50%	1/2
25%	1/4
20%	1/5
10%	1/10
5%	1/20





Percentage as Fractions

We can use these equivalences to find the value of other percentages.



▶▶ Example: What is 30% of 40?

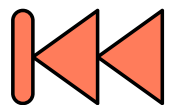
We can get 10% of 40 first since it is a factor of 30%.

Note that: 10% = 1/10

$$10\% \text{ of } 40 = \frac{1}{10} \times 40 = 4$$

Then, we can multiply the result by 3 to get the 30% of 40.

$$30\% \text{ of } 40 = 3 \times \left(\frac{1}{10} \times 40 \right) = 3 \times 4 = 12$$



Finding Percentage Directly

We can also find the percentage of a number directly by expressing the percentage as a fraction or decimal and multiplying it by the given quantity.

▶▶ Example: What is 12% of 50?

- 1 Express the percentage as a decimal.

$$12\% = 0.12$$

- 2 Multiply the result to the given quantity.

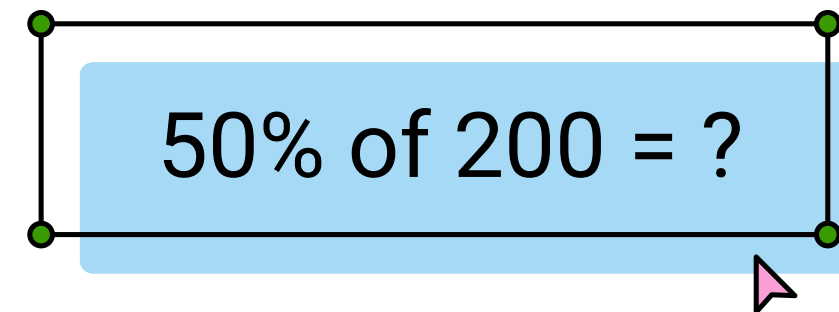
$$0.12 \times 50 = 6$$

Finding Percentage Directly

Let's solve.

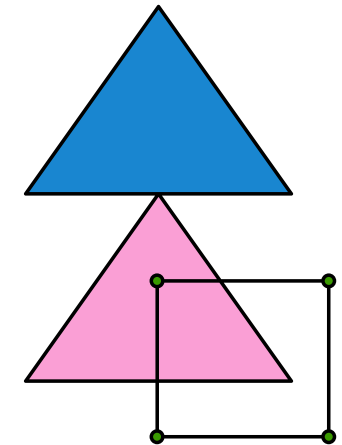
- ▶▶ 50% of Year 6 students are boys.
If there are 200 Year 6 students, how many are boys?

▶▶ Solution:

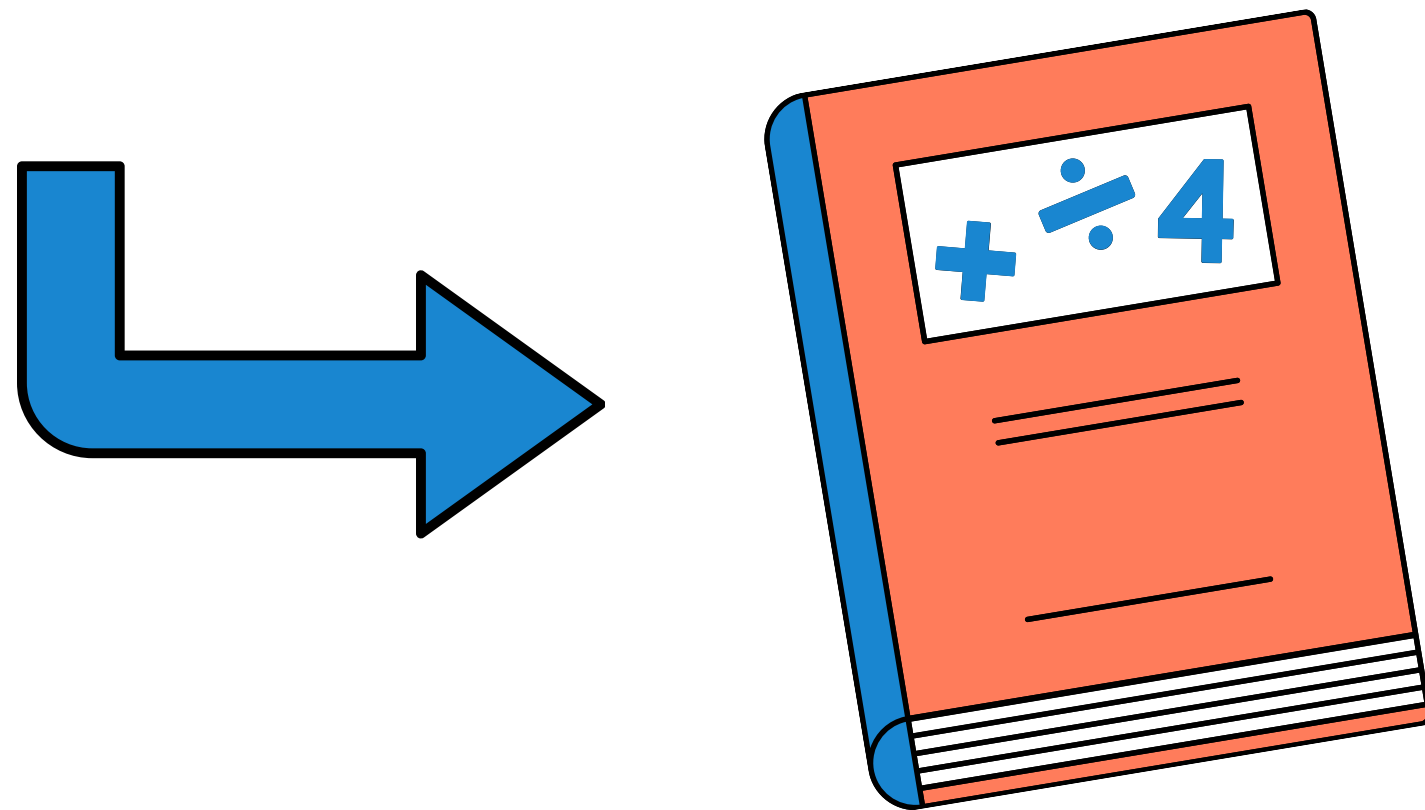

$$50\% \text{ of } 200 = ?$$

$$50\% \text{ of } 200 = 0.5 \times 200 = 100$$

100 Year 6 students are boys.



Math Marathon



▶▶ Let's solve each item!

How fast can you find the percentage of these quantities?

1 50% of 60 =

2 25% of 32 =

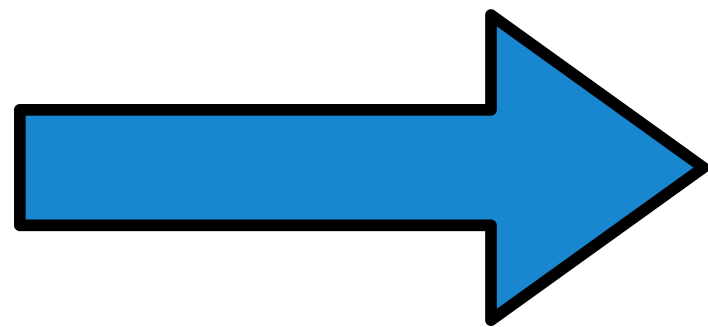
3 40% of 95 =

4 67% of 200 =

5 34% of 50 =

Math Marathon

▶▶ Answer Key



Tip: You can also switch
the values through the
commutative property!

Example:

$$50\% \text{ of } 60 = 60\% \text{ of } 50$$

▶▶ Let's solve each item!

How fast can you find the percentage of
these quantities?

1 50% of 60 $= \frac{1}{2} \times 60 = 30$

2 25% of 32 $= \frac{1}{4} \times 32 = 8$

3 40% of 95 $= \frac{2}{5} \times 95 = 38$

4 67% of 200 $= 0.67 \times 200 = 134$

5 34% of 50 $= 0.34 \times 50 = 17$

QUESTION #1

What is 25% written as a fraction?

- A. $\frac{1}{2}$
- B. $\frac{1}{4}$
- C. $\frac{3}{4}$
- D. $\frac{2}{5}$





QUESTION #2

Which of these is the same as
50%?

- A. 0.05%
- B. 0.005%
- C. 5%
- D. 0.5%

QUESTION #3

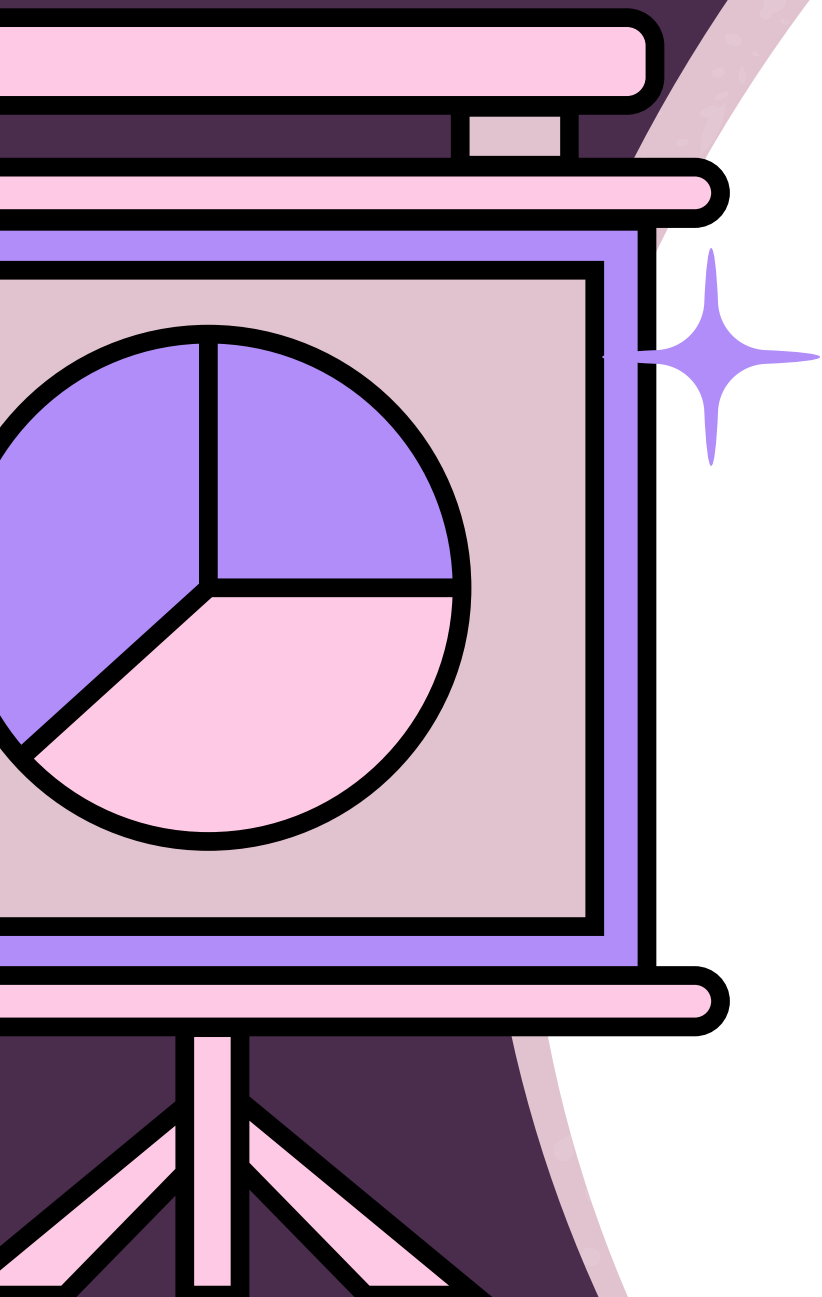
A shirt costs ₱500. It is on sale for 20% off. How much is the discount?

- A. ₱80
- B. ₱100
- C. ₱150
- D. ₱120

QUESTION #4

**Ella got 18 out of 20 on a test.
What percentage did she get?**

- A. 80%**
- B. 85%**
- C. 90%**
- D. 95%**



QUESTION #5

A battery shows 75% charged. What fraction of the battery is charged?

A. $\frac{1}{4}$

B. $\frac{3}{4}$

C. $\frac{1}{2}$

D. $\frac{2}{5}$

QUESTION #6

Which is greater?

- A. 50% of 40
- B. 25% of 100
- C. 10% of 200
- D. 60% of 30





QUESTION #7

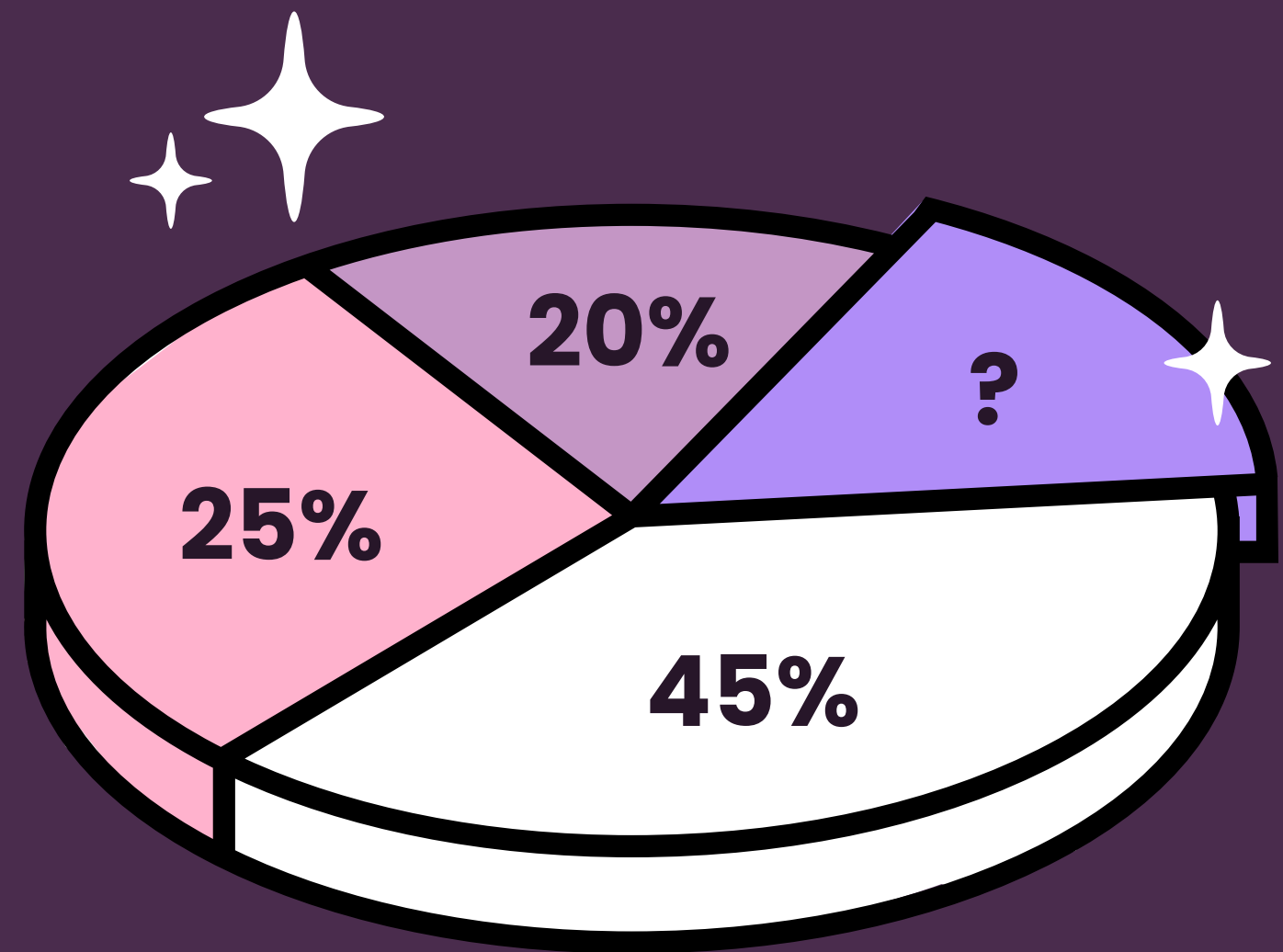
Liam spends 40% of his day at school. How many hours is that?

- A. 6.4 hours
- B. 4 hours
- C. 8 hours
- D. 5.5 hours

QUESTION #8

Look at this pie chart. What percentage does the question mark represent?

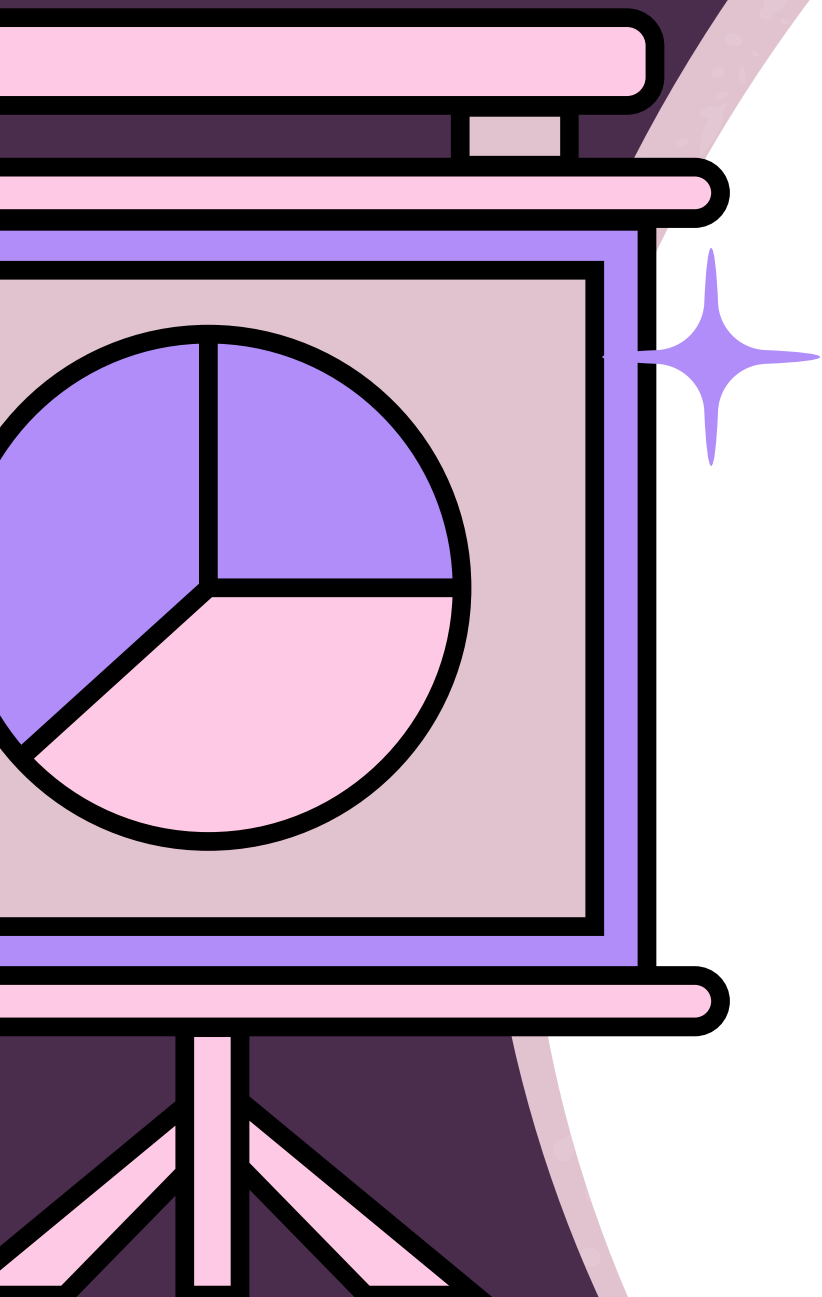
- A. 25%
- B. 50%
- C. 75%
- D. 10%



QUESTION #9

If 60% of the class are girls and there are 30 students, how many are girls?

- A. 18**
- B. 16**
- C. 12**
- D. 20**



QUESTION #10

A toy is ₱1,000 and has a 10% discount. What is the final price?

- A. ₱950**
- B. ₱900**
- C. ₱850**
- D. ₱1,100**

ANSWER KEY



1. B

2. D

3. A

4. C

5. B

6. A

7. A

8. D

9. A

10. A



THANK YOU

